

The electronic component of cosmic rays as an ionizing agent of the intergalactic medium during the epoch of reionization

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Abstract

Cosmic rays accelerated by the first stellar populations are injected into a predominantly neutral intergalactic medium (IGM) at the same epoch during which that medium is being reionized. We follow the kinetic energy K_e of a cosmic-ray electron injected at $z_i = 20$ and $z_i = 10$ down to $z = 5.5$, integrating simultaneously the seven relevant loss channels — adiabatic expansion, synchrotron, inverse Compton on the cosmic microwave background (CMB) with the exact Klein–Nishina kernel, Coulomb scattering on the residual plasma, collisional excitation and collisional ionization of neutral hydrogen, and bremsstrahlung. We show that the (z, K_e) plane partitions into four regimes separated by boundaries that we compute rather than assume: $K_1 = 13.6$ eV, below which only Coulomb heating operates; $K_2 = 7.6 \times 10^4$ – 1.8×10^5 eV, where collisional ionization gives way to inverse Compton; $K_3 = 4.2$ – 7.9 MeV, above which the up-scattered CMB photons become ionizing; and $K_4 = 3.2$ – 4.3×10^8 eV, above which those photons free-stream because their mean free path exceeds the Hubble radius. Solving the ionization cascade self-consistently — every generation of the secondary shower integrated along its own trajectory, with the energy-dependent binary-encounter-dipole secondary spectrum — we find that the yield Y per injected electron is strongly non-linear in K_{ini} : the derived $W = K_{\text{ini}}/Y$ has a broad minimum of 35.4 eV near 1 keV, where the deposition fractions $(f_{\text{ion}}, f_{\text{exc}}, f_{\text{heat}}) = (0.38, 0.42, 0.19)$ reproduce the $x_e \rightarrow 0$ Monte Carlo limit, but rises to 208 eV at 1 MeV and $\sim 10^8$ eV at 1 TeV, so no constant W describes the yield. Y flattens at 7.7×10^3 ($z_i = 20$) and 1.2×10^4 ($z_i = 10$) for $K_{\text{ini}} \gtrsim 10^8$ eV: above K_4 every additional electron volt is radiated into photons that leave the medium. The four boundaries divide the leptonic contribution into two productive bands separated by a dead one: direct collisional ionization below K_2 , a band $K_2 < K_e < K_3$ in which inverse Compton dominates but the up-scattered photons fall below the ionization threshold and the energy is lost to

the sub-ionizing background, a second productive band $K_3 < K_e < K_4$ in which those photons are ionizing and are absorbed locally, and outright escape above K_4 . Because a diffusive-shock-acceleration spectrum places most of its energy above K_4 , the leptonic component is an intrinsically inefficient reionizing agent unless the injection spectrum is soft.

Key words: intergalactic medium – dark ages, reionization, first stars – cosmic rays – radiation mechanisms: non-thermal

1 Introduction

1.1 The epoch of reionization

The epoch of reionization (EoR) is the last global phase transition of the baryonic Universe: the moment at which the intergalactic hydrogen left neutral by cosmological recombination was returned to an ionized state by the first luminous sources. Its end point is now pinned down with some precision. The distribution of effective optical depths in the Lyman- α forest of the XQR-30 quasar sample requires that reionization-related fluctuations — in the ultraviolet background, in the residual neutral fraction, or in the IGM temperature — persist until at least $z = 5.3$, so that the process cannot have finished appreciably earlier [Bosman et al., 2022]. At the other end, the Thomson optical depth to the CMB measured by *Planck*, $\tau = 0.054 \pm 0.007$, places the midpoint of a rapid transition near $z \simeq 7.7$ and disfavors a substantial ionized fraction before $z \sim 15$ [Planck Collaboration et al., 2020]. The interval $5.5 \lesssim z \lesssim 20$ adopted throughout this work therefore brackets the entire transition together with the preceding era in which the first sources were already operating.

What is far less settled is the *photon budget*: which sources supplied the ionizing photons, and whether they supplied enough of them at the right time. Star-forming galaxies are the standard answer [Robertson, 2022], but the *James Webb Space Telescope* has complicated it

rather than closed it. The abundance and ionizing efficiency ξ_{ion} inferred for early galaxies are high enough that, taken at face value with canonical escape fractions, they would complete reionization too early and overshoot both the CMB and the Lyman- α forest constraints — a tension now referred to as the photon budget crisis [Muñoz et al., 2024]. A budget that is simultaneously over-supplied in photons and under-constrained in timing is precisely the situation in which secondary, non-stellar ionizing channels deserve to be quantified rather than assumed negligible. The purpose of this paper is to quantify one such channel from first principles.

1.2 Cosmic dawn

Reionization does not begin with reionization. It begins with cosmic dawn, the interval during which the first collapsed structures light up and start depositing energy in a medium that is still almost fully neutral. The observational handle on this era is the redshifted 21-cm line, and it has recently become quantitative: the HERA Phase I power-spectrum limits require that the IGM was heated above the adiabatic cooling floor no later than $z = 10.4$, which rules out the whole family of “cold reionization” histories in which the gas remains at its adiabatic temperature until ionization fronts arrive [HERA Collaboration et al., 2023]. Something was already injecting energy into the neutral IGM well before the ionized bubbles overlapped.

That early energy injection is what makes cosmic dawn the natural starting point of the present analysis. Whatever heats the IGM at $z \sim 10$ – 20 must do so by depositing the kinetic energy of some non-thermal particle population into a gas of density $n_{\text{HI}} \simeq 2.6 \times 10^{-4}$ – $1.8 \times 10^{-3} \text{ cm}^{-3}$, and the same deposition that raises the temperature will, above the appropriate threshold, also produce free electrons. The heating and ionization channels are not independent: they are two branches of the same energy-loss problem, and they are set by the same cross-sections. The HERA constraint tells us the heating branch was active; the question this paper addresses is how much of the same energy budget could have gone into the ionization branch.

1.3 The first stellar populations

The sources responsible are the first stellar populations. Metal-free Pop III stars form in minihaloes of $\sim 10^5$ – $10^6 M_{\odot}$ from $z \sim 30$ onwards; primordial gas is strongly susceptible to fragmentation, so they form in small clusters with a roughly logarithmically flat mass function extending to several tens of solar masses, and feedback from the most massive members regulates all subsequent star formation [Klessen and Glover, 2023]. These stars are short-lived, and a large fraction of them end as core-collapse supernovae within a few million years of forma-

tion — on a timescale much shorter than the $\sim 290 \text{ Myr}$ that separates $z = 20$ from $z = 10$ in the adopted cosmology.

The relevance of this population to the present problem is that it delivers, in a single stellar generation, all four of the astrophysical engines known to accelerate particles to relativistic energies: fast radiatively driven and rotationally driven stellar winds, supernova blast waves, accreting compact objects in high-mass X-ray binaries (HMXBs), and the magnetized rotators left behind by the explosions. Each of these operates in an environment that is poorer in metals and, plausibly, richer in magnetic energy per unit mass than its low-redshift counterpart, and each of them terminates in a shock or a magnetosphere in which charged particles are accelerated out of the thermal pool. The first stellar populations are, in other words, the first cosmic-ray sources.

1.4 Winds, supernovae, HMXBs and compact remnants

Of these engines the supernova blast wave is the best characterized. Particle acceleration at non-relativistic quasi-parallel collisionless shocks proceeds through diffusive shock acceleration (DSA), and kinetic particle-in-cell simulations now show that protons *and* electrons are accelerated simultaneously at such shocks, both developing power-law momentum distributions with the universal DSA index -4 , corresponding to $dN/dK \propto K^{-2}$ in the relativistic limit [Park et al., 2015]. The same simulations, together with the observed $\sim 1\%$ electron fraction of the local Galactic cosmic-ray flux near a GeV, motivate an electron-to-proton number ratio $K_{e/p} \sim 10^{-2}$ at injection. High-mass X-ray binaries contribute along a different route: their X-ray output scales inversely with metallicity, so the metal-poor systems characteristic of early galaxies are expected to be substantially more luminous per unit star formation than local analogues [Kaur et al., 2022], and the same accretion-powered outflows that produce that X-ray emission are efficient particle accelerators. Taken together these engines make early star-forming galaxies copious cosmic-ray sources [Owen et al., 2017, Ruszkowski and Pfrommer, 2023].

The crucial step for what follows is escape. A cosmic ray produced inside a minihalo only matters for the IGM if it leaves the halo, and low-mass haloes are poorly equipped to confine it: their escape velocities and magnetic confinement times are small, and the photoevaporation that accompanies the first ionizing sources removes much of the confining gas. Low-energy cosmic rays produced by Pop III supernova remnants are therefore expected to be released more or less directly into the surrounding neutral IGM, and this is the configuration assumed by essentially all subsequent treatments of cosmic-ray feedback at high redshift [Sazonov and Sun-

yaev, 2015, Leite et al., 2017, Jana and Nath, 2018]. We adopt it here: the electron whose history we integrate is one that has already escaped its host halo and finds itself in the diffuse IGM at redshift z_i .

1.5 Cosmic rays in the early intergalactic medium

The idea that this escaped population might matter cosmologically is not new. Nath and Biermann [1993] asked directly whether cosmic rays from young galaxies could reionize the IGM, and the question has been revisited repeatedly since, with progressively more complete microphysics [Samui et al., 2005, Sazonov and Sunyaev, 2015, Leite et al., 2017, Jana and Nath, 2018, Bera et al., 2023, Gessey-Jones et al., 2023]. The modern consensus is sharply asymmetric between the two branches identified above. On the heating side the effect is real and substantial: cosmic rays from primordial supernovae can raise the IGM above the CMB temperature by $z \sim 15$ (Pop III) or $z \sim 10$ (Pop II) [Sazonov and Sunyaev, 2015], and detailed transport calculations give $\Delta T \simeq 10\text{--}200$ K by $z = 10$ depending on the injection spectrum [Leite et al., 2017]. Because the heating is short-ranged and clustered around star-forming sites it also leaves a distinctive small-scale signature in the 21-cm power spectrum that distinguishes it from X-ray heating [Gessey-Jones et al., 2023], although for typical diffusion coefficients the heating becomes effectively uniform at high redshift [Jana and Nath, 2018].

On the ionization side the same literature concludes that the contribution is negligible [Leite et al., 2017]. That conclusion, however, has almost invariably been reached for the *hadronic* component, and for good reason: cosmic-ray protons carry the bulk of the energy budget, and their Coulomb and ionization losses in a neutral medium are the natural first thing to compute. The electronic component has received far less attention, even though it interacts with the medium through a completely different set of channels — inverse Compton scattering on the CMB in particular, which at fixed total energy is suppressed for protons by $(m_e/m_p)^4 \simeq 10^{-13}$ and is therefore irrelevant, but which for electrons at these redshifts dominates everything else above ~ 100 keV. Whether the negligible-ionization verdict carries over to the leptonic component is not obvious *a priori*, and it is the question we set out to answer.

1.6 Scope: the leptonic component

We therefore restrict this paper to the electronic component of the cosmic-ray flux and defer the hadronic branch entirely. This is a deliberate separation rather than an omission. Cosmic-ray protons at these energies lose energy through Coulomb collisions and hadronic pp interactions, the latter producing pions whose decay in-

jects secondary $e^\pm$ and γ -rays; those secondaries re-enter the same leptonic cascade we analyse here, so the per-electron yields computed below are exactly the quantities a subsequent hadronic calculation would need as input. Separating the two also keeps the uncertainties separate: the leptonic problem is governed by cross-sections that are known to better than a few per cent, whereas the hadronic branch carries the additional uncertainty of the pion multiplicity and the proton spectrum. The hadronic contribution will be treated in a companion paper.

Within this scope our strategy is to compute everything *per injected electron*, as a function of the injection energy K_{ini} and the injection redshift z_i , and to postpone any convolution with a source spectrum to the very end. The per-electron yield is a property of the IGM and of atomic physics alone; the source spectrum is a property of the accelerator, and it is by far the more uncertain of the two. Presenting the results in this order means that the central results of this paper — the boundaries of the four regimes and the ionization yield curve — remain valid regardless of what is eventually assumed about the accelerators, and that a reader with a different spectrum in mind can re-weight our curves without repeating the integration.

2 Energy losses of an electron in the intergalactic medium

2.1 Fiducial medium and the seven loss channels

We adopt a spatially uniform, pure-hydrogen IGM with $n_{\text{HI}}(z) = 1.8 \times 10^3 [(1+z)/21]^3 \text{ m}^{-3}$ (that is, 1.8×10^{-3} , 2.6×10^{-4} and $5.3 \times 10^{-5} \text{ cm}^{-3}$ at $z = 20, 10$ and 5.5), a residual ionized fraction $x_e = n_e/n_{\text{HI}} = 10^{-4}$, and a magnetic field frozen into the expansion, $B(z) = B_0(1+z)^2$ with $B_0 = 1 \text{ nG}$ comoving. Cosmological quantities use the *Planck* 2018 parameters [Planck Collaboration et al., 2020] as tabulated in *astropy* [Astropy Collaboration et al., 2022]; in this cosmology $z = 20, 10$ and 5.5 correspond to cosmic ages of 178.1, 471.4 and 1038.5 Myr. Helium is neglected, which is conservative for the ionization yields quoted below.

Seven loss channels act on the electron simultaneously, and we integrate them together rather than in the more usual sequential approximation. Adiabatic losses follow $\dot{K} = -H(z)p^2c^2/E$. Synchrotron and inverse Compton losses are treated in the same isotropic-pitch-angle form, $\dot{K} = -\frac{4}{3}\sigma_{\text{T}}cU(p^2c^2/E_0^2)$, with $U = U_B$ and $U = U_{\text{CMB}}F_{\text{KN}}(b)$ respectively, where F_{KN} is the exact Klein–Nishina kernel integrated over the Planck spectrum [Blumenthal and Gould, 1970] and $b = 4\gamma kT_{\text{CMB}}(z)/E_0$. Coulomb losses on the residual plasma follow Gould [1972] with maximum fractional transfer $\delta = 0.5$. Colli-

sional excitation $H(1s) \rightarrow H(np)$ is summed over $n = 2-10$ using the scaled plane-wave Born cross-sections of Stone et al. [2002], and collisional ionization uses the relativistic binary-encounter-Bethe (RBEB) cross-section of Kim et al. [2000], with the energy lost per ionizing collision taken as the binding energy plus the mean secondary energy. Bremsstrahlung on neutral and ionized hydrogen follows Blumenthal and Gould [1970] with a continuous interpolation between the screened and unscreened limits.

2.2 The magnetic field is dynamically irrelevant

One consequence of the fiducial choice deserves to be stated at the outset, because it simplifies the entire analysis. Since both U_B and U_{CMB} scale as $(1+z)^4$, their ratio is a redshift-independent constant, and for $B_0 = 1$ nG it is $U_B/U_{\text{CMB}} = 9.5 \times 10^{-8}$. Synchrotron losses are therefore about seven orders of magnitude below inverse Compton losses at *every* redshift and *every* energy in our window; equipartition with the CMB would require $B_0 \simeq 3.2$ μG comoving, four orders of magnitude above the fiducial value. The synchrotron channel is retained in the integration for completeness but never affects any result reported here, and this is a physical statement about the early IGM rather than a numerical artefact.

The magnetic field nevertheless matters enormously for *transport*, and in the opposite direction. The gyro-radius $r_g = p_\perp/(eB)$ at $z = 10$, where the proper field is 121 nG, is 1.3×10^{-8} pc for a 1 MeV electron and only 9×10^{-3} pc even at 1 TeV. Compared with the tens of comoving megaparsecs of path length that the same electron traverses before thermalizing (Sect. 8), the electron is tied to its field line to one part in 10^{10} . Any statement about *where* the ionizations occur is therefore a statement about cosmic-ray transport, not about the loss physics; we return to this point in Sect. 8.

2.3 Integration

The seven rates are assembled into a single stiff ordinary differential equation for $K_e(t)$, integrated with an implicit Radau scheme from the cosmic time corresponding to z_i down to $z = 5.5$, with a terminal event that stops the integration when the electron reaches the thermal floor $K_{\text{term}} = \frac{3}{2}kT_{\text{CMB}}(z)$. To obtain the energy budget of each channel separately we integrate an augmented eight-state system that carries the seven cumulative integrals $\int L_i dt$ alongside K_e ; this guarantees that the channel budgets are evaluated on the true trajectory rather than on an approximate one, and that they sum to $K_{\text{ini}} - K_e(t)$ by construction. The ionization yield of Sect. 5 is obtained from two further augmented states carrying the primary collision count $\int n_{\text{HI}} v \sigma_{\text{ion}} dt$ and

the cascade total of Eq. (1), so that the number of ionizations and the energy lost to them are computed from the same cross-section on the same trajectory.

Figure 1 shows the resulting loss rates at $z = 20$ and $z = 10$. The structure is the same at both redshifts and is what organizes the rest of this paper: a Coulomb-dominated regime at the lowest energies, a broad plateau in which collisional ionization of neutral hydrogen carries the losses, and a steeply rising inverse-Compton regime above it in which the total rate grows as K_e^2 . Adiabatic, synchrotron and bremsstrahlung losses never dominate anywhere in the plane; the peak fractional contribution of the adiabatic term over the whole window is $\lesssim 0.09$, reached for $K_{\text{ini}} \sim 10^6$ eV at $z_i = 10$ (Fig. 3).

3 The four-regime partition of the (z, K_e) plane

3.1 Boundaries computed, not assumed

The conceptual scheme underlying this work divides cosmic-ray electrons into four classes — very cold, cold, warm and hot — on the basis of which deposition channel their energy ultimately reaches. We can now replace those nominal divisions by boundaries computed from the loss physics itself. Figure 2 shows the dominant channel as a function of (z, K_e) over the full window $5.5 \leq z \leq 20$, with four boundary curves overlaid, and Table 1 lists their values. The first, $K_1 = 13.61$ eV, is the hydrogen ionization threshold and is redshift-independent by construction; below it no inelastic atomic channel is open and Coulomb scattering on the residual plasma is the only sink. The second, K_2 , is the crossover at which the inverse-Compton rate overtakes the collisional-ionization rate; it runs from 7.6×10^4 eV at $z = 20$ to 1.8×10^5 eV at $z = 5.5$, bracketing the nominal 100 keV division. Its mild redshift dependence has a transparent origin: collisional losses scale as $n_{\text{HI}} \propto (1+z)^3$ while inverse Compton scales as $U_{\text{CMB}} \propto (1+z)^4$, so the crossover moves to lower energy as z increases.

The remaining two boundaries are not properties of the electron’s cooling at all but of the *fate of the photons it produces*, and they are the reason the inverse-Compton regime must be subdivided. K_3 is the electron energy at which the up-scattered CMB photons first reach the hydrogen ionization threshold of 13.6 eV; evaluating the exact head-on scattering kinematics against the high-energy tail of the Planck spectrum (the 99th percentile of the CMB energy density, $x = h\nu/kT = 8.17$) gives $K_3 = 4.20, 5.99$ and 7.93 MeV at $z = 20, 10$ and 5.5 . Using instead the mean CMB photon, $x = 2.70$, raises these to $7.7, 10.8$ and 14.2 MeV, so the threshold is of order a few MeV with a factor-of-two definitional spread. K_4 is the electron energy above which those photons

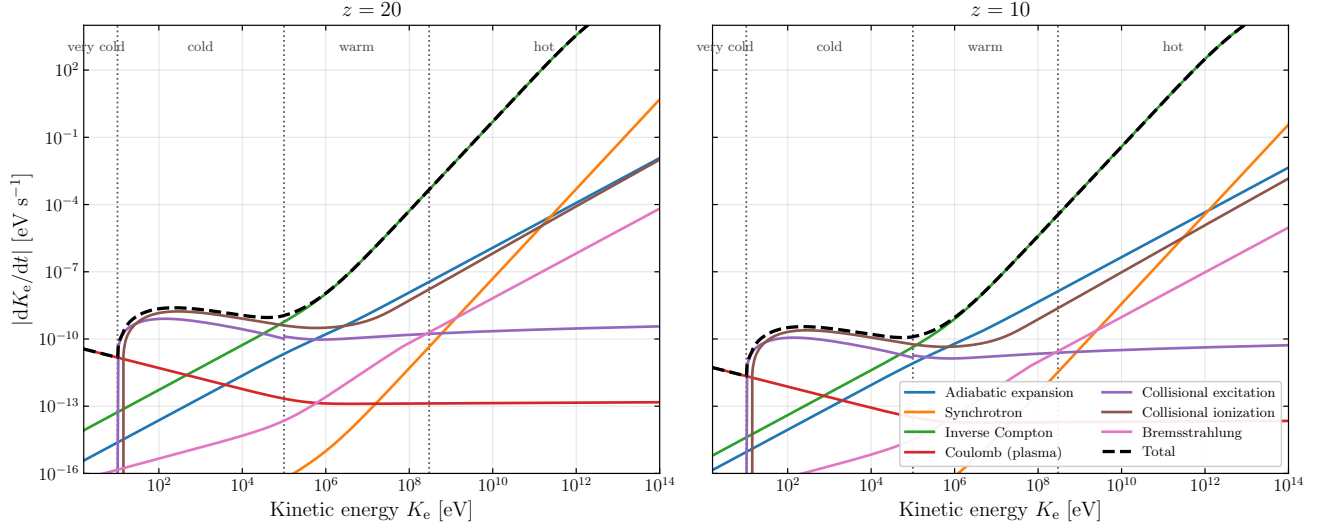


Figure 1: Energy-loss rate $|dK_e/dt|$ of a cosmic-ray electron in the fiducial IGM, decomposed into the seven channels, at $z = 20$ (left) and $z = 10$ (right). The dotted vertical lines mark the nominal boundaries of the four regimes of the conceptual scheme (10.2 eV, 100 keV, 300 MeV); the boundaries computed in Sect. 3 are given in Table 1. Synchrotron losses lie ~ 7 orders of magnitude below inverse Compton at all energies because $U_B/U_{\text{CMB}} = 9.5 \times 10^{-8}$ for $B_0 = 1$ nG comoving.

Table 1: Computed boundaries of the four regimes, and the photon energy E_γ^{free} at which the proper mean free path equals the Hubble radius $c/H(z)$. All electron energies in eV.

z	K_1	K_2	K_3	E_γ^{free}	K_4
20.0	13.61	7.60×10^4	4.20×10^6	1820	3.25×10^8
15.0	13.61	9.32×10^4	4.88×10^6	1550	3.43×10^8
10.0	13.61	1.23×10^5	5.99×10^6	1272	3.75×10^8
8.0	13.61	1.43×10^5	6.67×10^6	1151	3.94×10^8
5.5	13.61	1.82×10^5	7.93×10^6	983	4.29×10^8

escape, and it is derived in Sect. 7.

3.2 Where the energy actually goes

Knowing which channel is fastest at a given instant is not the same as knowing where an electron’s energy ends up, because the electron sweeps downwards through the regimes as it cools. Figure 3 therefore shows the integrated budget: the fraction of K_{ini} that each channel has absorbed by the time the electron reaches the thermal floor. The picture it gives is cleaner than Fig. 1 might suggest. For $K_{\text{ini}} \lesssim 10^4$ eV the budget is essentially independent of both K_{ini} and z_i and is shared between collisional ionization ($\simeq 0.48$ – 0.72) and collisional excitation ($\simeq 0.26$ – 0.40), with Coulomb heating taking $\simeq 0.12$ at 100 eV and falling rapidly. For $K_{\text{ini}} \gtrsim 10^7$ eV inverse Compton takes everything: $f_{\text{IC}} = 0.93$ at 10 MeV and > 0.99 above 100 MeV ($z_i = 10$).

The transition between these two limits is narrow, and

it is where the physically interesting behaviour lives. At $K_{\text{ini}} = 10^5$ eV and $z_i = 10$ the split is $f_{\text{ion}} = 0.61$, $f_{\text{exc}} = 0.18$, $f_{\text{IC}} = 0.17$; one decade higher, at 10^6 eV, it has already inverted to $f_{\text{IC}} = 0.66$, $f_{\text{ion}} = 0.19$. Two decades in injection energy therefore convert the electron from a device that deposits two-thirds of its energy directly into the atomic gas to one that radiates two-thirds of it away. Everything that follows about the ionizing efficiency of cosmic-ray electrons is a consequence of this inversion.

4 Very cold electrons: Coulomb heating

4.1 The only open channel below 10.2 eV

Below the Lyman- α excitation threshold at 10.204 eV no inelastic channel involving neutral hydrogen is kinematically accessible: the electron cannot excite an atom, cannot ionize one, and radiates negligibly. The only remaining sink is elastic and quasi-elastic Coulomb scattering off the residual free electrons of the partially ionized plasma, which converts the cosmic-ray electron’s kinetic energy directly into thermal energy of the gas [Gould, 1972]. This is visible in Fig. 1 as the red curve that rises towards low energies and is the unique surviving channel below K_1 , and in Fig. 2 as the red band at the bottom of the plane. Between 10.2 and 13.6 eV collisional excitation opens and takes over, so the strictly Coulomb-only regime is bounded above by the Lyman- α threshold, exactly as in the conceptual scheme.

The efficiency of this channel is set by the residual

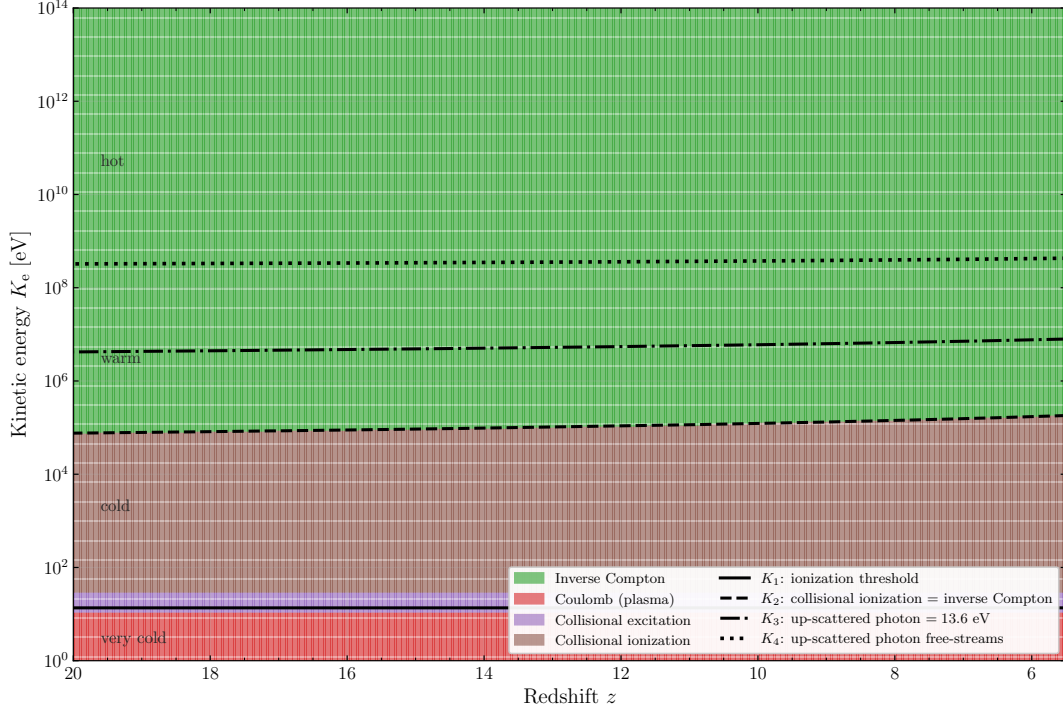


Figure 2: Dominant energy-loss channel in the (z, K_e) plane over $5.5 \leq z \leq 20$ (shaded), with the four computed boundaries of Table 1 overlaid: K_1 the ionization threshold, K_2 the collisional-ionization/inverse-Compton crossover, K_3 the energy at which up-scattered CMB photons become ionizing, and K_4 the energy above which those photons free-stream. K_3 subdivides the warm regime rather than bounding it.

ionization fraction, and this is the one place where the fiducial $x_e = 10^{-4}$ enters a result directly. The Coulomb rate is proportional to $n_e = x_e n_{\text{HI}}$, so a medium ten times more ionized heats ten times faster at fixed cosmic-ray energy. In the integrated budget (Fig. 3) Coulomb losses account for $\simeq 12\%$ of a 100 eV electron’s energy but only 1.8×10^{-3} of a 10 keV one, because a fast electron spends very little of its life at the low energies where Coulomb scattering competes. Very cold electrons are thus not a by-product of the cascade of fast ones; they matter only if the source injects them directly, which is precisely the regime in which cosmic-ray heating of the early IGM has been found to be significant [Sazonov and Sunyaev, 2015, Leite et al., 2017].

4.2 Thermal response: heating up, recombining less

The consequence of that heating is twofold, and the two effects reinforce each other. The first is direct: raising the IGM kinetic temperature raises the 21-cm spin temperature above the CMB temperature and switches the global signal from absorption to emission. This is the effect that current cosmic-ray calculations were designed to capture, and it is substantial — $\Delta T \simeq 10$ –200 K by $z = 10$ for plausible injection spectra [Leite et al., 2017], enough to satisfy the HERA requirement

that the gas be above the adiabatic floor by $z = 10.4$ [HERA Collaboration et al., 2023] without invoking X-ray binaries. Because the heating is deposited close to its sources it is also spatially structured, which enhances small-scale 21-cm power relative to X-ray heating models [Gessey-Jones et al., 2023], although for the diffusion coefficients usually assumed the structure washes out and the heating is effectively uniform at high redshift [Jana and Nath, 2018].

The second consequence is indirect but bears directly on the reionization budget. The case-B recombination coefficient of hydrogen scales approximately as $\alpha_B \propto T^{-0.7}$, so a medium heated from the adiabatic floor of a few kelvin to 10^2 – 10^4 K recombines more slowly, and each ionizing photon produced by any source — stellar or otherwise — goes further. The magnitude of this effect must be stated carefully: cosmic rays heat the neutral gas, whereas recombinations occur in the H II regions, which photoionization already holds near 10^4 K, so an extra $\Delta T \lesssim 200$ K changes α_B there by $\lesssim 1.4\%$; the larger lever is the pre-heating of the gas *before* it is ionized, discussed in Sect. 9.4. Measured IGM temperatures at the tail end of reionization, $T_0 \simeq 1.0$ – 1.2×10^4 K at $z = 5.4$ – 5.8 [Gaikwad et al., 2020], confirm that the gas is indeed hot by then. Cosmic-ray heating therefore contributes to the ionization budget through a channel

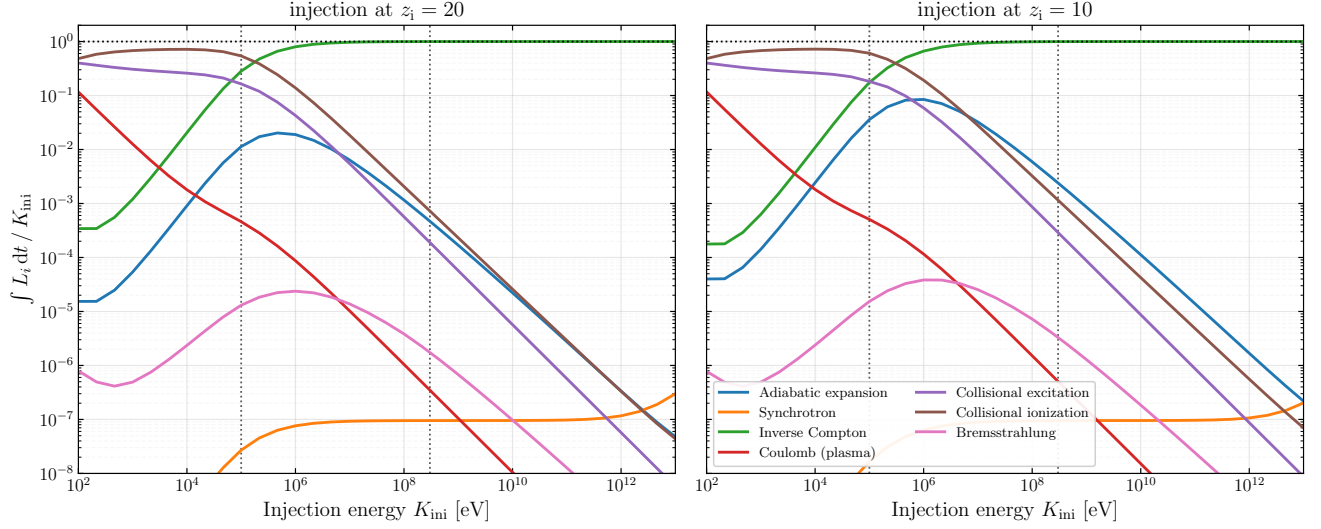


Figure 3: Fraction of the injected kinetic energy absorbed by each channel, $\int L_i dt / K_{\text{ini}}$, integrated along the true trajectory from injection to thermalization, for $z_i = 20$ (left) and $z_i = 10$ (right). Dotted vertical lines as in Fig. 1.

that is entirely distinct from producing free electrons itself: by suppressing the sink term. This is worth separating explicitly, because it means the total cosmic-ray effect on reionization is not simply the number of ionizations the cosmic rays themselves cause.

5 Cold electrons: direct collisional ionization

5.1 The ionization yield per injected electron

Between K_1 and K_2 collisional ionization of neutral hydrogen is the dominant loss channel, and the electron deposits its energy by making free electrons directly. Counting them requires following not only the primary but every generation it spawns, because each ionizing collision ejects a secondary of energy ϵ and any secondary with $\epsilon > 13.6$ eV ionizes in turn. We therefore solve the fixed point

$$\frac{dY}{dt} = n_{\text{HI}} v \sigma_{\text{ion}}(K_e) \left[1 + \langle Y(\epsilon, z) \rangle_{p(\epsilon|K_e)} \right], \quad (1)$$

marching upward in energy (a secondary always has $\epsilon \leq (K_e - B)/2 < K_e$, so the recursion is causal) and integrating *every* generation along its own cosmological trajectory with the same seven-channel loss model. No ceiling is placed on the secondary energy: a secondary born above K_4 loses its energy to escaping photons exactly as a primary does.

The secondary spectrum $p(\epsilon|K_e)$ is the BEB singly-differential cross section [Kim and Rudd, 1994] rebuilt from the same dipole coefficients that generate the total

RBEB cross-section of Kim et al. [2000], so that the differential and integral descriptions are one and the same physics; for hydrogen it gives $N_i = 0.43431$, and integrating it reproduces the total cross-section to 0.01% below 300 eV and 2.6% at 10 keV. This matters because the spectrum is energy dependent: the mean secondary energy runs from 1.5 eV at $K_e = 20$ eV to 9.7, 21.8 and 40.4 eV at 100 eV, 1 keV and 100 keV, so successive generations are launched into different parts of the yield curve, and $\langle Y(\epsilon) \rangle \neq Y(\langle \epsilon \rangle)$ wherever Y is non-linear.

5.2 The yield is not proportional to the injected energy

Figure 4 and Table 2 give the result. The shower is not a small correction: it multiplies the yield by 1.02 at 100 eV, 1.25 at 1 keV, 1.57 at 12 keV and $\simeq 2.0$ above 10 MeV, reaching 2.3–2.5 at TeV energies, and it is the generations beyond the first that carry the agreement with the Monte Carlo literature. Writing the derived quantity $W(K_{\text{ini}}) = K_{\text{ini}}/Y$ makes the point that no single number reproduces the yield: W falls from 40.1 eV at 100 eV to a broad minimum of 35.4 eV near 1 keV, is back to 52.8 eV at 110 keV, and then rises without limit — 208 eV at 1 MeV, 1.5×10^3 at 10 MeV, 1.4×10^8 at 1 TeV. Any calculation that adopts $Y = K_{\text{ini}}/W$ with a constant W is therefore wrong by orders of magnitude for $K_{\text{ini}} \gtrsim 1$ MeV, and the error is in the non-conservative direction.

The minimum is worth isolating, because it is where our calculation makes contact with the literature. At the minimum the deposition fractions are $f_{\text{ion}} = 0.38$, $f_{\text{exc}} = 0.42$ and $f_{\text{heat}} = 0.19$, and $W = 35.4$ eV at both $z_i = 20$ and $z_i = 10$ to three significant figures — the $x_e \rightarrow 0$

Table 2: Cascade ionization yield Y per injected electron, the primary-only count N_1 for comparison, and the derived $W = K_{\text{ini}}/Y$. Y includes every generation of the shower, each integrated along its own trajectory.

K_{ini} [eV]	z_i	N_1	Y	$W = K_{\text{ini}}/Y$ [eV]
1.1×10^2	20	2.60	2.65	40.1
1.0×10^3	20	22.6	28.2	35.4
1.2×10^4	20	206	323	37.1
1.1×10^5	20	1163	2089	52.8
1.0×10^6	20	2568	4880	208
8.6×10^7	20	3276	6667	1.30×10^4
1.1×10^{12}	20	3293	7648	1.42×10^8
1.1×10^2	10	2.60	2.65	40.1
1.0×10^3	10	22.6	28.3	35.4
1.2×10^4	10	208	326	36.7
1.1×10^5	10	1307	2361	46.8
1.0×10^6	10	3499	6755	151
8.6×10^7	10	4894	10327	8.36×10^3
1.1×10^{12}	10	4930	12369	8.77×10^7

limit of Shull and van Steenberg [1985] and Furlanetto and Stoeve [2010], recovered here from atomic cross-sections alone with no fitted parameter. That agreement also disposes of the value of 20.8 eV sometimes quoted as a rule of thumb: it would require $f_{\text{ion}} = B/20.8 = 0.65$, far above the $\simeq 0.39$ ceiling a neutral medium allows, and in a neutral IGM the Lyman-band photons produced by the excitation channel cannot ionize H(1s) at all. It coincides instead with $B + \langle \epsilon \rangle$ at $K_e = 61$ eV, that is with the energy a slow primary loses per ionizing collision — a stopping-power quantity, not a deposition W -value.

5.3 Saturation: the ceiling on the yield

The most consequential feature of Fig. 4 is not the near-linear low-energy regime but its termination. Above $K_{\text{ini}} \simeq 10^8$ eV the yield flattens to $Y \simeq 7.7 \times 10^3$ for $z_i = 20$ and 1.2×10^4 for $z_i = 10$; increasing the injection energy from 10^8 to 10^{12} eV — four orders of magnitude — adds only 15 and 20% respectively, the residual creep coming from the handful of very energetic secondaries that each start a shower of their own. Equivalently, W rises from 35 eV to $\sim 10^8$ eV across the same range: a TeV cosmic-ray electron is, per unit energy, a factor 2×10^6 less efficient at ionizing hydrogen than a 1 keV one.

The origin of the ceiling is the competition already visible in Fig. 1. Above K_2 the inverse-Compton rate exceeds the collisional-ionization rate and grows as K_e^2 , so the more energy the electron carries the faster it is stripped by the CMB and the less time it spends in the energy range where it can ionize atoms. The ceiling is therefore set not by K_{ini} but by the number of ionizations accumulated during the final passage through the cold regime, which is nearly the same for every electron

that enters it from above. That the ceiling is lower at $z_i = 20$ than at $z_i = 10$ follows from the same argument: at higher redshift U_{CMB} is larger by $(21/11)^4 \simeq 13$, cooling is correspondingly faster, and the electron passes through the productive window more quickly.

6 Warm electrons: Compton-mediated ionization

6.1 From electron energy to photon energy

Above K_2 the electron no longer ionizes the medium directly in any significant measure; it up-scatters CMB photons, and whether those photons ionize anything depends entirely on where they land in energy. This makes the warm regime qualitatively different from the cold one: the deposition is non-local, mediated by radiation, and it has an energy threshold of its own. Between K_2 and K_3 the up-scattered photons remain below 13.6 eV and cannot ionize hydrogen at all. Photons emerging in the narrow band between 10.2 and 13.6 eV are Lyman-series photons and do useful work — they couple the 21-cm spin temperature to the gas through the Wouthuysen–Field effect — but photons below 10.2 eV simply redshift away without interacting. A significant part of the inverse-Compton budget in the range $10^5 \lesssim K_{\text{ini}} \lesssim 10^{6.5}$ eV is therefore lost to the radiation background rather than deposited in the gas.

Above $K_3 \simeq 4\text{--}8$ MeV the situation changes: the up-scattered photons are ionizing, and each one that is absorbed liberates a photoelectron which then re-enters the cascade analysed in Sect. 5 at its own energy. The total ionization count for a warm electron is consequently $Y^{\text{tot}} = N_\gamma + \sum_j Y(K_j)$, where N_γ counts the primary photoionizations and the sum runs over the photoelectrons, each contributing according to the same yield curve Y established above. Our integration follows the collisional cascade but not this photon-mediated branch, so the yields plotted in Fig. 4 are, in the warm regime, strictly a lower bound; the photon-mediated contribution is the subject of the follow-up calculation described in Sect. 9.3.

6.2 Absorption is local

What makes this channel viable at all is that the photons produced between K_3 and K_4 are absorbed essentially where they are made. The left panel of Fig. 5 shows the proper mean free path of a photon in the fiducial IGM, computed from the exact hydrogen photoionization cross-section [Karzas and Latter, 1961] together with the Klein–Nishina cross-section on bound and free electrons. At $z = 10$ a photon at the ionization threshold travels 2×10^{-4} Mpc and one at 100 eV trav-

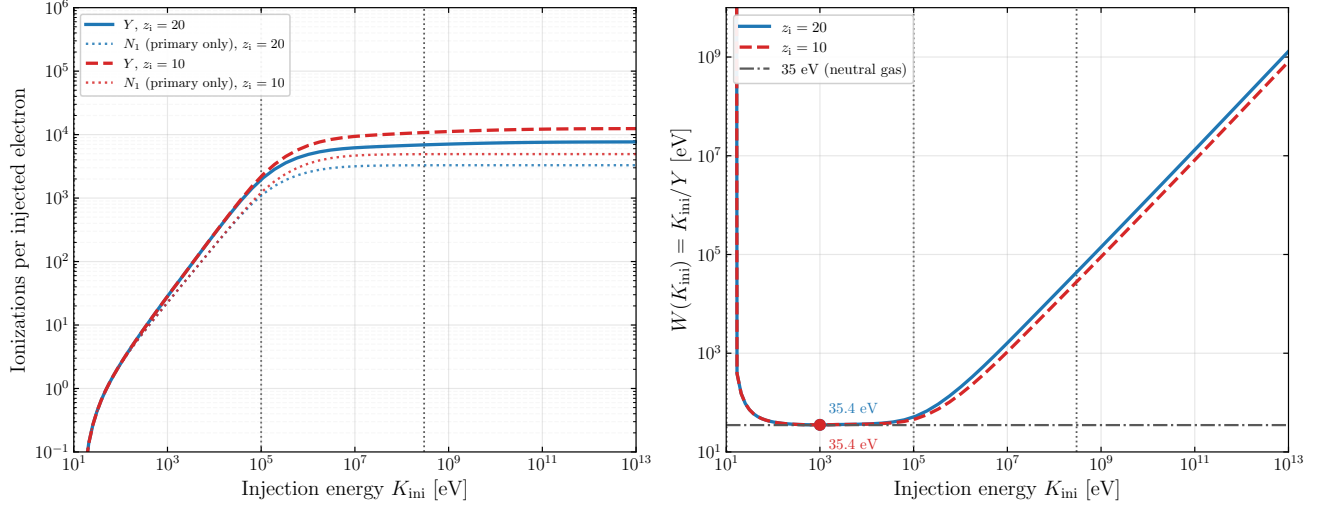


Figure 4: Left: ionizations produced by one injected electron. Solid and dashed curves give the cascade yield Y , which includes every generation of the secondary shower; dotted curves give the primary-only count N_1 for comparison. Right: the derived $W(K_{\text{ini}}) = K_{\text{ini}}/Y$, with its minimum marked. W is manifestly not a constant — it has a broad minimum of 35.4 eV near 1 keV and rises without limit above $\sim 10^5$ eV — so the yield cannot be written as K_{ini} divided by a fixed energy per ionization. Dotted vertical lines mark the nominal regime boundaries.

els 0.08 Mpc, both utterly negligible compared with the 217 Mpc Hubble radius. The steep E^{-3} decline of the photoionization cross-section means that the medium remains highly opaque throughout the band that matters.

The consequence is that the warm regime behaves, to a good approximation, like the cold one with an extra step inserted: energy leaves the electron as radiation but is redeposited within a distance that is negligible on any cosmological scale, so the ionizations still occur in the immediate neighbourhood of the source. This is what justifies treating the warm and cold contributions on the same footing when they are later summed into a shell budget (Sect. 8), and it is what fails — abruptly — in the hot regime.

7 Hot electrons: energy lost to infinity

7.1 The free-streaming threshold

The hydrogen photoionization cross-section falls as E_γ^{-3} , so the opacity of the IGM collapses within a single decade above the threshold, and at some photon energy the mean free path exceeds the Hubble radius and the photon ceases to interact with the medium at all. Solving $\lambda(E_\gamma) = c/H(z)$ with the full opacity gives $E_\gamma^{\text{free}} = 1.82, 1.27$ and 0.98 keV at $z = 20, 10$ and 5.5 (Fig. 5, left, and Table 1). The threshold moves to lower energy as the Universe expands because the density drops faster than the Hubble rate. This is the same transparency window that governs the energy deposition

efficiency of exotic injections in the dark ages [Slatyer et al., 2009, Kanzaki and Kawasaki, 2008, Slatyer, 2016], and its location near 1 keV is a robust feature of a neutral hydrogen-dominated medium.

Translating this back into an electron energy through the Compton kinematics gives the fourth boundary: $K_4 = 3.25 \times 10^8, 3.75 \times 10^8$ and 4.29×10^8 eV at $z = 20, 10$ and 5.5 . Electrons above roughly 300 MeV convert their energy into photons that leave the medium without depositing anything — they are, in the language of the conceptual scheme, lost to infinity. It is worth noting that this boundary is remarkably flat in redshift, varying by only 32% across a factor of 3.2 in $(1+z)$, because the rise in E_γ^{free} with redshift is nearly compensated by the rise in the CMB photon energy that is being up-scattered.

7.2 The dominant population is the useless one

The astrophysical significance of K_4 is that it sits well below the energies at which cosmic-ray accelerators deposit most of their power. A DSA spectrum $dN/dK \propto K^{-p}$ with $p \simeq 2$ [Park et al., 2015, Ruszkowski and Pfrommer, 2023] distributes energy equally per logarithmic interval, so an accelerator operating from GeV to TeV places the overwhelming majority of its *energy* above K_4 even though most of its *particles* may lie below. For those electrons the ionization yield is not merely reduced, it is capped at the plateau of Sect. 5, and the entire excess is radiated into a background that the IGM cannot see.

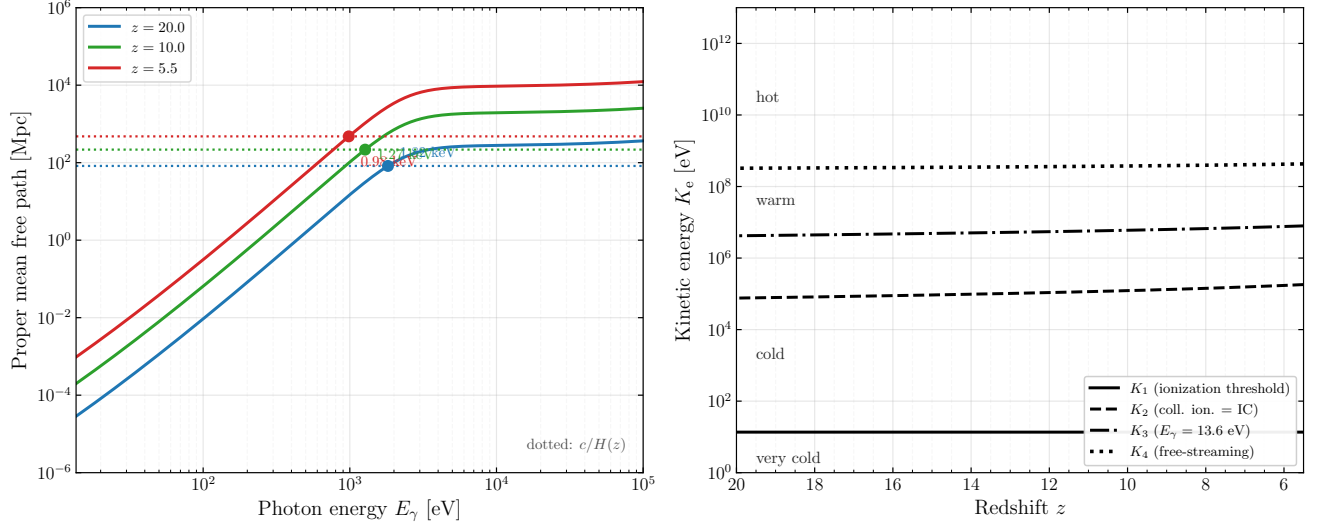


Figure 5: Left: proper mean free path of a photon in the fiducial IGM at $z = 20, 10$ and 5.5 (solid), including hydrogen photoionization [Karzas and Latter, 1961] and Klein–Nishina scattering; dotted lines mark the Hubble radius $c/H(z)$, and the filled circles locate E_γ^{free} , where the two are equal. Right: the resulting electron-energy boundaries K_1 – K_4 as functions of redshift.

This is the central negative result of the paper, and it is worth stating precisely, because the four boundaries do not carve out a single efficient window but two productive bands separated by a dead one. Below K_2 the electron ionizes directly at $W \simeq 35$ – 53 eV. Between K_2 and K_3 inverse Compton takes the energy but the up-scattered photons are below 13.6 eV: apart from the thin Lyman-band slice noted in Sect. 6, that energy is lost to the sub-ionizing radiation background, and this band is the least productive part of the whole plane. Between K_3 and K_4 the photons are ionizing and are reabsorbed within a negligible distance, so the energy returns to the gas as photoionizations — a contribution our direct-only integration does not count, which is why the plateau in Fig. 4 understates the true yield there. Above K_4 nothing is recovered at all. Since a DSA spectrum with $p \simeq 2$ places most of its energy above K_4 , any scenario proposing cosmic-ray electrons as a significant reionization source must explain not merely how enough energy is injected, but how it is kept out of the escaping band — either by a soft spectrum or by a cutoff below ~ 400 MeV. We return to this in Sect. 9.2.

8 Ionizations per spherical shell per Δz

8.1 Cooling time and the redshift interval

Having established the yield per electron, the next step of the scheme is to distribute those ionizations in space

and time — to convert $Y(K_{\text{ini}}, z_i)$ into a number of ionizations per spherical shell per redshift interval Δz . The temporal part is straightforward, because the integration already returns the redshift at which each electron thermalizes. Figure 6 shows K_e against $\Delta z = z_i - z$ and makes the timescales explicit. At $z_i = 10$, an electron injected at 10^5 eV thermalizes by $\Delta z = 0.40$, one at 10^6 eV by $\Delta z = 1.82$, and one at 10^{12} eV by $\Delta z = 2.85$. At $z_i = 20$ the same energies thermalize by $\Delta z = 0.25, 0.89$ and 1.23 respectively.

Two features of these numbers matter for the shell construction. First, the thermalization redshift saturates with injection energy for exactly the reason the yield does: above K_4 the extra energy is dumped into escaping photons on a timescale that is short compared with the Hubble time, so a TeV electron and a 100 MeV electron reach the thermal floor at almost the same epoch. Second, cooling is markedly faster at high redshift, so the deposition is more sharply localized in Δz at $z = 20$ than at $z = 10$. A shell budget constructed with a fixed Δz will therefore resolve the injection history well at early times and poorly at late times; the natural choice is a Δz that tracks the cooling time, which for our purposes means $\Delta z \lesssim 0.3$ at $z \gtrsim 15$ and $\Delta z \lesssim 1$ at $z \sim 10$.

8.2 The radial extent is set by transport, not by cooling

The spatial part is much less straightforward, and it is important to be explicit about what our calculation does and does not determine. Figure 7 shows the kinetic

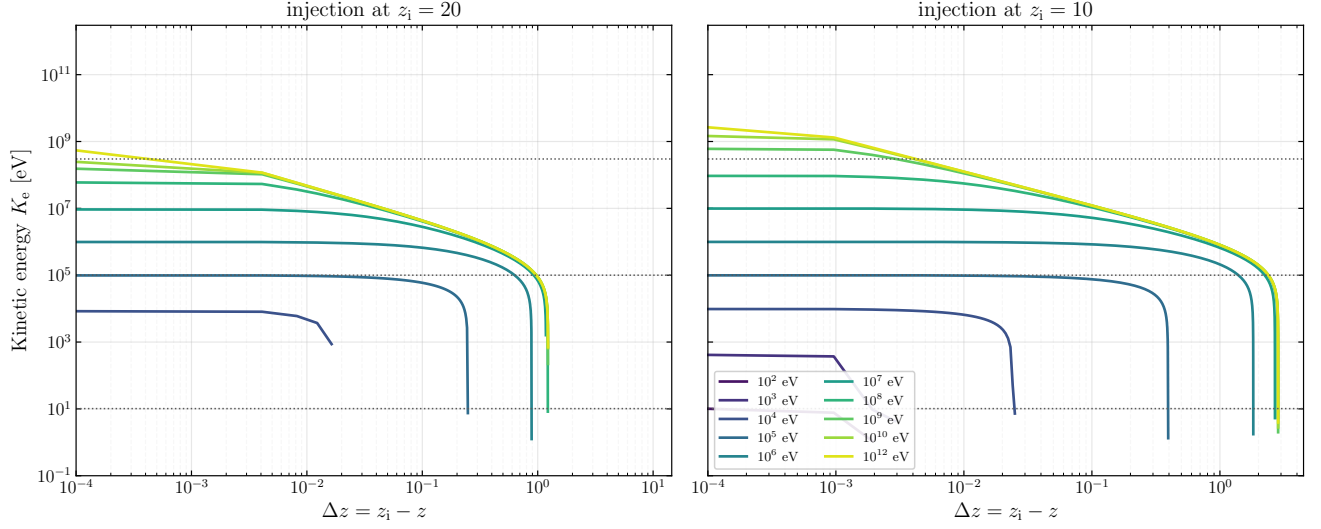


Figure 6: Kinetic energy against elapsed redshift interval $\Delta z = z_i - z$ for electrons injected at $z_i = 20$ (left) and $z_i = 10$ (right), for injection energies from 10^2 to 10^{12} eV. Curves terminate at the thermal floor $\frac{3}{2}kT_{\text{CMB}}(z)$. Dotted horizontal lines mark the nominal regime boundaries.

energy against the comoving path length $\chi = \int v(1+z) dt$ accumulated along the trajectory. The path length saturates in the same way the other quantities do: an electron injected at 10^5 eV traverses 8.0 comoving Mpc before thermalizing, one at 10^6 eV traverses 49.6 Mpc, and everything above 10^8 eV traverses 78.1 Mpc, against a comoving light-travel distance of 2757 Mpc from $z = 20$ to $z = 5.5$.

These path lengths are emphatically *not* shell radii. As noted in Sect. 2, the gyroradius in the fiducial field is 10^{-8} – 10^{-2} pc, some ten orders of magnitude below the path length, so the electron does not travel in a straight line: it spirals along a field line and its displacement from the injection site is governed by the cross-field diffusion coefficient, not by how far it has flown. The correct statement is that the path length is a rigorous *upper bound* on the shell radius, saturating at ~ 78 comoving Mpc, and that the actual radius requires a transport model. This is precisely the quantity that controls whether cosmic-ray heating is patchy or uniform: Jana and Nath [2018] find that inhomogeneous deposition requires $D \lesssim 10^{26} \text{ cm}^2 \text{ s}^{-1}$ at $z \sim 10$, and that for typical values the heating is uniform. Because the ionization we compute is deposited by the same particles, the same criterion applies to it, and we adopt the uniform-deposition limit in what follows.

9 Discussion

9.1 Estimating the global contribution

The global ionization budget follows from the per-electron yield by weighting it with the injection spec-

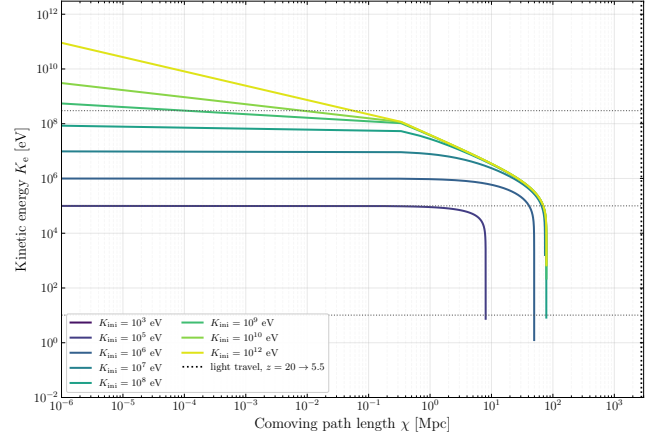


Figure 7: Kinetic energy against comoving path length $\chi = \int v(1+z) dt$ for electrons injected at $z_i = 20$. The path length saturates near 78 comoving Mpc for all $K_{\text{ini}} \gtrsim 10^8$ eV. Because the gyroradius is smaller by ten orders of magnitude, this is an upper bound on the displacement, not the displacement itself.

trum. Writing the number of electrons injected per unit comoving volume per unit energy as $dn/dK_{\text{ini}} = A K_{\text{ini}}^{-p}$ between K_{min} and K_{max} , the number of ionizations per unit comoving volume is $n_{\text{ion}} = \int A K_{\text{ini}}^{-p} Y(K_{\text{ini}}) dK_{\text{ini}}$, and the useful figure of merit is the spectrum-averaged energy per ionization

$$\overline{W}(p) = \frac{\int_{K_{\text{min}}}^{K_{\text{max}}} K_{\text{ini}}^{1-p} dK_{\text{ini}}}{\int_{K_{\text{min}}}^{K_{\text{max}}} K_{\text{ini}}^{-p} Y(K_{\text{ini}}) dK_{\text{ini}}}, \quad (2)$$

which converts a cosmic-ray electron energy density directly into an ionization density. The shape of $Y(K_{\text{ini}})$ established in Sect. 5 — close to linear up to $\sim 10^5$ eV, then bending over and flattening above $\sim 10^8$ eV — makes the behaviour of Eq. (2) easy to anticipate. For $p > 2$ both integrals are dominated by K_{min} , the flattening is never sampled, and $\bar{W}$ tends to the low-energy plateau of $W(K_{\text{ini}})$, $\simeq 37$ eV. For $p < 2$ the numerator is dominated by K_{max} while the denominator saturates, so $\bar{W}$ grows as K_{max}^{2-p} . The canonical DSA value $p = 2$ sits exactly at the transition, where $\bar{W}$ grows only logarithmically in the dynamic range of the accelerator.

Table 3 evaluates Eq. (2) numerically on the yield curves of Fig. 4, taking $K_{\text{min}} = 100$ eV and $K_{\text{max}} = 1$ TeV. The anticipated behaviour is borne out with some force. Between $p = 2.5$ and $p = 1.8$ — a range comfortably inside the spread of spectral indices that real accelerators produce — the spectrum-averaged energy per ionization degrades by a factor of 21, from 37.6 to 804 eV, and the comoving cosmic-ray electron energy density required to supply one ionization per hydrogen atom rises correspondingly from 1.2×10^{-17} to 2.5×10^{-16} erg cm $^{-3}$. The dependence on z_i is negligible by comparison, below 11% across the whole table. The global contribution of cosmic-ray electrons to reionization is therefore controlled almost entirely by the softness of the injection spectrum, and hardly at all by K_{max} , by the injection redshift, or by the total energy above a GeV. This inverts the usual intuition that more energetic accelerators are more effective, and it explains why the hadronic studies cited in Sect. 1 reached a negative verdict on ionization while reaching a positive one on heating: the proton spectra they adopt place their energy at exactly the energies where the leptonic yield saturates. Two caveats should be attached to Table 3. It counts collisional ionization only, so it still omits the photon-mediated ionizations of the band $K_3 < K_{\text{ini}} < K_4$ (Sect. 6); since the integrals for $p < 2$ are dominated by precisely that band, the $p = 1.8$ entry should be read as a conservative worst case rather than an estimate. It also requires only one ionization per atom, neglecting recombinations, so the energy densities quoted are floors. The photon-mediated warm-regime contribution of Sect. 6 and a full recombination bookkeeping are deferred to the companion paper.

9.2 Which scenarios are worth exploring

Equation (2) is, read differently, a design specification for the accelerator. To make cosmic-ray electrons a competitive ionizing source one needs an injection spectrum that is steep ($p \gtrsim 2.2$), that extends down to K_{min} of order tens of eV rather than being cut off at MeV energies, and that operates at a redshift where the surrounding medium is still neutral enough to absorb the resulting photons. The first two requirements point away from

Table 3: Spectrum-averaged energy per ionization $\bar{W}$ from Eq. (2) for an injection spectrum $dn/dK_{\text{ini}} \propto K_{\text{ini}}^{-p}$ with $K_{\text{min}} = 100$ eV and $K_{\text{max}} = 1$ TeV, and the comoving cosmic-ray electron energy density $\varepsilon_{\text{CR}} = n_{\text{H}} \bar{W}$ needed to supply one ionization per hydrogen atom ($n_{\text{H}} = 1.94 \times 10^{-7}$ cm $^{-3}$ comoving).

p	$z_i = 20$		$z_i = 10$	
	$\bar{W}$ [eV]	ε_{CR} [erg cm $^{-3}$]	$\bar{W}$ [eV]	ε_{CR} [erg cm $^{-3}$]
1.8	888	2.8×10^{-16}	804	2.5×10^{-16}
2.0	107	3.3×10^{-17}	102	3.2×10^{-17}
2.2	46.3	1.4×10^{-17}	45.3	1.4×10^{-17}
2.5	37.8	1.2×10^{-17}	37.6	1.2×10^{-17}
3.0	37.4	1.2×10^{-17}	37.4	1.2×10^{-17}

strong supernova blast waves — which are efficient precisely because they produce hard spectra extending to high energy — and towards weaker shocks, stellar wind termination shocks, and the reacceleration of an existing low-energy population. The third points to the earlier part of our window.

The redshift dependence works in two opposing directions and this is what makes the scenario question non-trivial. Injecting earlier means a denser, more neutral medium, a shorter cooling time and more sharply localized deposition (Fig. 6), all of which help; but it also means a $(1+z)^4$ -larger CMB energy density, which strengthens the inverse-Compton sink that steals energy from the ionization channel, and a lower saturation ceiling (7.7×10^3 versus 1.2×10^4 ionizations per electron between $z_i = 20$ and $z_i = 10$). The net effect is that the productive window in K_{ini} narrows with increasing redshift: K_2 falls from 1.8×10^5 to 7.6×10^4 eV between $z = 5.5$ and $z = 20$ (Table 1). Scenarios that inject at $z \sim 10$ –15 with a soft spectrum are therefore favoured over both earlier and harder alternatives, and this is a prediction that a source-population model can be tested against.

9.3 Comparison with the reionization history: is it worth exploring?

Set against the observed reionization history, the honest answer is a qualified yes, for reasons that have more to do with the thermal history than with the photon budget. Our results confirm, from the leptonic side and with the full loss physics, the negative verdict that the hadronic literature reached on ionization [Leite et al., 2017]: the flattening of Y above ~ 100 MeV caps the yield of exactly the particles that carry the accelerator’s energy, and no plausible normalization of a hard spectrum turns 10^4 ionizations per electron into a competitive share of the 1.9×10^{-7} cm $^{-3}$ (comoving) of hydrogen that must be ionized. Cosmic-ray electrons are not going to reion-

ize the Universe on their own.

What the channel does deliver is a thermal signature at redshifts where the 21-cm experiments are now sensitive, and the same energy budget that makes the ionization contribution negligible fixes the two quantities together. Because every fast electron, whatever its injection energy, deposits what it does deposit through the *same* cold-regime cascade, the heat released per ionization produced is nearly invariant: our channel-resolved integration of the cascade gives 6.0–9.6 eV of heat per ionization over $K_{\text{ini}} = 10^2\text{--}10^6$ eV, and the value is identical at $z_i = 10$ and $z_i = 20$ to three significant figures, because the inverse-Compton leak removes heat and ionizations in the same proportion and cancels in the ratio. Converting the heat into a temperature rise of a gas with 1.081 particles per hydrogen atom, this fixes the number of cosmic-ray ionizations per hydrogen atom to

$$\frac{N_{\text{ion}}}{n_{\text{H}}} = \frac{\Delta T}{(4.3\text{--}6.9) \times 10^4 \text{ K}}, \quad (3)$$

independently of the injection spectrum and of redshift. Equation (3) is the result that makes the channel testable: a 21-cm measurement of ΔT is simultaneously a measurement of the leptonic contribution to reionization.

9.4 Does the leptonic channel narrow or widen the photon budget crisis?

The photon budget crisis is the statement that the directly observed galaxy population already produces too many ionizing photons too early [Muñoz et al., 2024], so a resolution must remove ionizations from the early Universe or delay them. Equation (3) says the leptonic channel does the opposite, and it says by how much. Taking the heating range that the hadronic calculations obtain for plausible injection spectra, $\Delta T = 10\text{--}200$ K at $z = 10$ [Leite et al., 2017] — borrowed here as a scale, not as a leptonic prediction — Eq. (3), evaluated at the $K_{\text{ini}} \gtrsim 1$ keV end where a cosmic-ray spectrum carries its energy, gives 2.3×10^{-4} to 4.6×10^{-3} ionizations per hydrogen atom, i.e. 0.02–0.5% of reionization, and a Thomson-depth increment $\delta\tau = 3.5 \times 10^{-5}\text{--}7.7 \times 10^{-4}$, at most a tenth of the *Planck* uncertainty $\sigma(\tau) = 0.007$ [Planck Collaboration et al., 2020]. Those numbers are consistent with the 21-cm limits that bracket the same epoch: the LOFAR upper limits of $\Delta_{21}^2 < (54.3 \text{ mK})^2$ at $z \simeq 9.1$ [Mertens et al., 2025] disfavour, at 95% credibility, only IGM states with mean gas temperature below 44 K and with less than 46% of the volume ionized or heated [Ghara et al., 2025], so a few tens of kelvin of cosmic-ray heating is permitted but not required, while the HERA requirement that the gas exceed the adiabatic floor $T_{\text{ad}} = 2.70$ K by $z = 10.4$ [HERA Collaboration et al., 2023] is comfortably met at the upper end of the range.

Every other route by which the heat could act on the reionization history runs in the same direction. Raising the temperature of ionized gas lowers the case-B recombination coefficient (Sect. 4.2), which removes sink term and therefore accelerates reionization; pre-heating the neutral gas also smooths it in pressure and lowers the recombination-weighted clumping factor [Pawlik et al., 2009], pushing against the high value $C = 6.2_{-2.1}^{+4.1}$ that resolves the crisis in the Austin et al. [2025] analysis of 1721 galaxies at $5.6 < z < 6.5$. The one mechanism that would run the other way — raising the filtering mass so that star formation in minihaloes is suppressed — does not apply to the population at issue: 100 K of heating gives a Jeans mass $M_{\text{J}} = 1.5 \times 10^6 M_{\odot}$ at $z = 10$, within the halo range $4.4 \times 10^5\text{--}1.1 \times 10^7 M_{\odot}$ that the SARAS 3 non-detection already disfavors [Bevins et al., 2022], but some three decades below the $\gtrsim 10^9 M_{\odot}$ hosts of the $M_{\text{UV}} < -15$ galaxies whose directly measured output creates the tension. The leptonic channel therefore *widens* the crisis on every channel that acts on the observed population, by of order 10^{-3} of the discrepancy — which is to say it is observationally irrelevant to the photon budget and diagnostically valuable to the thermal history. That is the sense in which the channel is worth pursuing, and quantifying the joint heating-plus-ionization response against the $z = 5.3$ end point of Bosman et al. [2022], the *Planck* optical depth and the 21-cm limits simultaneously is the question we intend to answer next.

10 Summary

We have integrated the seven simultaneous energy-loss channels of a cosmic-ray electron in a neutral intergalactic medium from $z = 20$ and $z = 10$ down to $z = 5.5$, and used the result to quantify the ionizing efficiency of the leptonic component of cosmic rays during the epoch of reionization. Our main conclusions are:

1. The (z, K_e) plane partitions into four regimes with boundaries that follow from the loss physics: $K_1 = 13.6$ eV, $K_2 = 0.76\text{--}1.8 \times 10^5$ eV, $K_3 = 4.2\text{--}7.9$ MeV and $K_4 = 3.2\text{--}4.3 \times 10^8$ eV (Table 1).
2. Synchrotron losses are irrelevant at all energies and redshifts for $B_0 = 1$ nG comoving, since $U_B/U_{\text{CMB}} = 9.5 \times 10^{-8}$ independently of z . The magnetic field nevertheless dominates transport, the gyroradius being ten orders of magnitude below the cooling path length.
3. The ionization yield $Y(K_{\text{ini}}, z_i)$, obtained by solving the cascade over every generation of the shower, is not proportional to K_{ini} . The derived $W = K_{\text{ini}}/Y$ has a broad minimum of 35.4 eV near 1 keV — reproducing the $x_e \rightarrow 0$ limit of Shull and van Steenberg [1985] and Furlanetto and Stoever [2010] from

atomic cross-sections alone — and rises to 208 eV at 1 MeV and $\sim 10^8$ eV at 1 TeV. Y flattens at 7.7×10^3 ($z_i = 20$) and 1.2×10^4 ($z_i = 10$) ionizations per electron above 10^8 eV. The secondary shower contributes a factor 1.02–2.5, rising with K_{ini} .

4. Above K_4 the up-scattered CMB photons exceed $E_\gamma^{\text{free}} \simeq 1$ keV, at which the mean free path exceeds the Hubble radius, and their energy leaves the medium entirely.
5. The contribution is therefore split into two productive bands — $K_1 < K_e < K_2$ (direct collisional ionization) and $K_3 < K_e < K_4$ (photon-mediated, absorbed locally) — separated by a dead band $K_2 < K_e < K_3$ in which the up-scattered photons are sub-ionizing.
6. The spectrum-averaged energy per ionization, Eq. (2), is controlled by the low-energy end of the injection spectrum for $p > 2$ and diverges as K_{max}^{2-p} for $p < 2$: soft spectra extending to low energies are the only competitive configuration.
7. Because all the deposited energy passes through the same cold-regime cascade, the heat released per ionization is nearly invariant (6.0–9.6 eV over $K_{\text{ini}} = 10^2$ – 10^6 eV, and independent of z_i to three significant figures). This locks the ionization contribution to the heating, Eq. (3), so that $\Delta T = 10$ – 200 K yields at most 4.6×10^{-3} ionizations per hydrogen atom and $\delta\tau \lesssim 7.7 \times 10^{-4}$. The leptonic channel thus *widens* rather than relieves the photon budget crisis of Muñoz et al. [2024], by a fraction of order 10^{-3} of the discrepancy, and is best regarded as a probe of the thermal history rather than a contributor to the ionizing budget.

Data and code availability

All curves shown were generated from a single shared physics module (`igm_losses.py`) and the figure script `make_paper_figures.py` accompanying this manuscript.

References

Astropy Collaboration, Adrian M. Price-Whelan, et al. The Astropy project: Sustaining and growing a community-oriented open-source project and the latest major release (v5.0) of the core package. *The Astrophysical Journal*, 935(2):167, 2022. doi: 10.3847/1538-4357/ac7c74. URL <https://arxiv.org/abs/2206.14220>.

D. Austin, T. Harvey, C. J. Conselice, N. J. Adams, V. Rusakov, Q. Li, et al. Resolving the ionizing photon budget crisis with JWST/NIRCam H II clumping constraints at $z \simeq 6$. *arXiv e-prints*, 2025. doi: 10.48550/arXiv.2512.10839. URL <https://arxiv.org/abs/2512.10839>.

Ankita Bera, Saumyadip Samui, and Kanan K. Datta. Impact of cosmic rays on the global 21-cm signal during cosmic dawn. *Monthly Notices of the Royal Astronomical Society*, 2023. doi: 10.1093/mnras/stac3814. URL <https://arxiv.org/abs/2202.12308>.

H. T. J. Bevins, A. Fialkov, E. de Lera Acedo, W. J. Handley, S. Singh, R. Subrahmanyan, and R. Barkana. Astrophysical constraints from the SARAS 3 non-detection of the cosmic dawn sky-averaged 21-cm signal. *Nature Astronomy*, 6:1473–1483, 2022. doi: 10.1038/s41550-022-01825-6. URL <https://arxiv.org/abs/2212.00464>.

George R. Blumenthal and Robert J. Gould. Bremsstrahlung, synchrotron radiation, and Compton scattering of high-energy electrons traversing dilute gases. *Reviews of Modern Physics*, 42(2):237–270, 1970. doi: 10.1103/RevModPhys.42.237. URL <https://ui.adsabs.harvard.edu/abs/1970RvMP...42..237B>.

Sarah E. I. Bosman, Frederick B. Davies, George D. Becker, Laura C. Keating, Rebecca L. Davies, Yongda Zhu, Anna-Christina Eilers, Valentina D’Odorico, et al. Hydrogen reionisation ends by $z = 5.3$: Lyman- α optical depth measured by the XQR-30 sample. *Monthly Notices of the Royal Astronomical Society*, 514(1):55–76, 2022. doi: 10.1093/mnras/stac1046. URL <https://arxiv.org/abs/2108.03699>.

Steven R. Furlanetto and Samuel Johnson Stoeve. Secondary ionization and heating by fast electrons. *Monthly Notices of the Royal Astronomical Society*, 404(4):1869–1878, 2010. doi: 10.1111/j.1365-2966.2010.16401.x. URL <https://arxiv.org/abs/0910.4410>.

Prakash Gaikwad, Michael Rauch, Martin G. Haehnelt, Ewald Puchwein, James S. Bolton, Laura C. Keating, Girish Kulkarni, et al. Probing the thermal state of the intergalactic medium at $z > 5$ with the transmission spikes in high-resolution Ly α forest spectra. *Monthly Notices of the Royal Astronomical Society*, 494(4):5091–5109, 2020. doi: 10.1093/mnras/staa907. URL <https://arxiv.org/abs/2001.10018>.

Thomas Gessey-Jones, Anastasia Fialkov, Eloy de Lera Acedo, Will J. Handley, and Renan Barkana. Signatures of cosmic ray heating in 21-cm observables. *Monthly Notices of*

- the Royal Astronomical Society*, 526(3):4262–4284, 2023. doi: 10.1093/mnras/stad2965. URL <https://arxiv.org/abs/2304.07201>.
- R. Ghara, S. Zaroubi, B. Ciardi, G. Mellema, S. K. Giri, F. G. Mertens, M. Mevius, L. V. E. Koopmans, I. T. Iliev, et al. Constraints on the state of the intergalactic medium at $z \sim 8 - 10$ using redshifted 21-cm observations with LOFAR. *Astronomy & Astrophysics*, 699:A109, 2025. doi: 10.1051/0004-6361/202554163. URL <https://arxiv.org/abs/2505.00373>.
- Robert J. Gould. Energy loss of fast electrons and positrons in a plasma. *Physica*, 60(1):145–154, 1972. doi: 10.1016/0031-8914(72)90227-3. URL <https://ui.adsabs.harvard.edu/abs/1972Phy....60..145G>.
- HERA Collaboration, Zara Abdurashidova, et al. Improved constraints on the 21 cm EoR power spectrum and the X-ray heating of the IGM with HERA Phase I observations. *The Astrophysical Journal*, 945(2):124, 2023. doi: 10.3847/1538-4357/acaf50. URL <https://arxiv.org/abs/2210.04912>.
- Ranita Jana and Biman B. Nath. Cosmic ray heating of intergalactic medium: patchy or uniform? *Monthly Notices of the Royal Astronomical Society*, 479(1):153–161, 2018. doi: 10.1093/mnras/sty1481. URL <https://arxiv.org/abs/1806.01295>.
- Toru Kanzaki and Masahiro Kawasaki. Electron and photon energy deposition in the Universe. *Physical Review D*, 78(10):103004, 2008. doi: 10.1103/PhysRevD.78.103004. URL <https://arxiv.org/abs/0805.3969>.
- W. J. Karzas and R. Latter. Electron radiative transitions in a Coulomb field. *The Astrophysical Journal Supplement Series*, 6:167–212, 1961. doi: 10.1086/190063. URL <https://ui.adsabs.harvard.edu/abs/1961ApJS....6..167K>.
- Harman Deep Kaur, Yuxiang Qin, Andrei Mesinger, Andrea Pallottini, Tassos Fragos, and Antara Basu-Zych. The 21-cm signal from the Cosmic Dawn: metallicity dependence of high mass X-ray binaries. *Monthly Notices of the Royal Astronomical Society*, 513(4):5097–5108, 2022. doi: 10.1093/mnras/stac1226. URL <https://arxiv.org/abs/2203.10851>.
- Yong-Ki Kim and M. Eugene Rudd. Binary-encounter-dipole model for electron-impact ionization. *Physical Review A*, 50(5):3954–3967, 1994. doi: 10.1103/PhysRevA.50.3954. URL <https://ui.adsabs.harvard.edu/abs/1994PhRvA..50.3954K>.
- Yong-Ki Kim, José Paulo Santos, and Fernando Pariente. Extension of the binary-encounter-dipole model to relativistic incident electrons. *Physical Review A*, 62(5):052710, 2000. doi: 10.1103/PhysRevA.62.052710. URL <https://journals.aps.org/prabstract/10.1103/PhysRevA.62.052710>.
- Ralf S. Klessen and Simon C. O. Glover. The first stars: Formation, properties, and impact. *Annual Review of Astronomy and Astrophysics*, 61:65–130, 2023. doi: 10.1146/annurev-astro-071221-053453. URL <https://arxiv.org/abs/2303.12500>.
- Natacha Leite, Carmelo Evoli, Marta D’Angelo, Benedetta Ciardi, Günter Sigl, and Andrea Ferrara. Do cosmic rays heat the early intergalactic medium? *Monthly Notices of the Royal Astronomical Society*, 469(1):416–424, 2017. doi: 10.1093/mnras/stx805. URL <https://arxiv.org/abs/1703.09337>.
- F. G. Mertens, M. Mevius, L. V. E. Koopmans, A. R. Offringa, S. Zaroubi, et al. Deeper multi-redshift upper limits on the epoch of reionization 21-cm signal power spectrum from LOFAR between $z = 8.3$ and $z = 10.1$. *Astronomy & Astrophysics*, 698:A186, 2025. doi: 10.1051/0004-6361/202554158. URL <https://arxiv.org/abs/2503.05576>.
- Julian B. Muñoz, Jordan Mirocha, John Chisholm, Steven R. Furlanetto, and Charlotte Mason. Reionization after JWST: a photon budget crisis? *Monthly Notices of the Royal Astronomical Society Letters*, 2024. doi: 10.1093/mnrasl/slae086. URL <https://arxiv.org/abs/2404.07250>.
- Biman B. Nath and Peter L. Biermann. Did cosmic rays reionize the intergalactic medium? *Monthly Notices of the Royal Astronomical Society*, 265(1):241–249, 1993. doi: 10.1093/mnras/265.1.241. URL <https://arxiv.org/abs/astro-ph/9306001>.
- Ellis R. Owen, Kinwah Wu, Pooja Surajbali, and Idunn B. Jacobsen. Cosmic rays in early star-forming galaxies and their effects on the interstellar medium. In *Proceedings of the 35th International Cosmic Ray Conference (ICRC2017)*, PoS, page 289, 2017. URL <https://arxiv.org/abs/1708.06808>.
- Jaehong Park, Damiano Caprioli, and Anatoly Spitkovsky. Simultaneous acceleration of protons and electrons at nonrelativistic quasiparallel collisionless shocks. *Physical Review Letters*, 114(8):085003, 2015. doi: 10.1103/PhysRevLett.114.085003. URL <https://arxiv.org/abs/1412.0672>.
- Andreas H. Pawlik, Joop Schaye, and Eveline van Scherpenzeel. Keeping the universe ionized: photoheating and the clumping factor of the high-redshift intergalactic medium. *Monthly Notices of the Royal Astronomical Society*, 394(4):1812–1824, 2009. doi: 10.1111/j.1365-2966.2009.14486.x. URL <https://arxiv.org/abs/0807.3963>.

- Planck Collaboration, N. Aghanim, Y. Akrami, et al. Planck 2018 results. vi. cosmological parameters. *Astronomy & Astrophysics*, 641:A6, 2020. doi: 10.1051/0004-6361/201833910. URL <https://arxiv.org/abs/1807.06209>.
- Brant E. Robertson. Galaxy formation and reionization: Key unknowns and expected breakthroughs by the James Webb Space Telescope. *Annual Review of Astronomy and Astrophysics*, 60:121–158, 2022. doi: 10.1146/annurev-astro-120221-044656. URL <https://arxiv.org/abs/2110.13160>.
- Mateusz Ruszkowski and Christoph Pfrommer. Cosmic ray feedback in galaxies and galaxy clusters. a pedagogical introduction and a topical review of the acceleration, transport, observables, and dynamical impact of cosmic rays. *The Astronomy and Astrophysics Review*, 31(1):4, 2023. doi: 10.1007/s00159-023-00149-2. URL <https://arxiv.org/abs/2306.03141>.
- Saumyadip Samui, Kandaswamy Subramanian, and Raghunathan Srianand. Can cosmic rays heat the intergalactic medium?, 2005. URL <https://arxiv.org/abs/astro-ph/0505590>. arXiv preprint.
- Sergey Sazonov and Rashid Sunyaev. Preheating of the Universe by cosmic rays from primordial supernovae at the beginning of cosmic reionization. *Monthly Notices of the Royal Astronomical Society*, 454(4):3464–3471, 2015. doi: 10.1093/mnras/stv2255. URL <https://ui.adsabs.harvard.edu/abs/2015MNRAS.454.3464S>.
- J. Michael Shull and Mark E. van Steenberg. X-ray secondary heating and ionization in quasar emission-line clouds. *The Astrophysical Journal*, 298:268–274, 1985. doi: 10.1086/163605. URL <https://ui.adsabs.harvard.edu/abs/1985ApJ...298..268S>.
- Tracy R. Slatyer. Indirect dark matter signatures in the cosmic dark ages. ii. ionization, heating, and photon production from arbitrary energy injections. *Physical Review D*, 93(2):023521, 2016. doi: 10.1103/PhysRevD.93.023521. URL <https://arxiv.org/abs/1506.03812>.
- Tracy R. Slatyer, Nikhil Padmanabhan, and Douglas P. Finkbeiner. CMB constraints on WIMP annihilation: Energy absorption during the recombination epoch. *Physical Review D*, 80(4):043526, 2009. doi: 10.1103/PhysRevD.80.043526. URL <https://arxiv.org/abs/0906.1197>.
- Philip M. Stone, Yong-Ki Kim, and Jean Paul Desclaux. Electron-impact cross sections for dipole- and spin-allowed excitations of hydrogen, helium, and lithium. *Journal of Research of the National Institute of Standards and Technology*, 107(4):327–337, 2002. doi: 10.6028/jres.107.026. URL <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC4859265/>.